A Common Electrical-Energy Basis for Comparing Lunar Oxygen and Propellant Production Routes

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Code, data, and all figures are reproducible from the open-source model at https://github.com/dubthree/lunar-propellant-energy-model.

Abstract

Lunar in-situ propellant production is widely proposed as the key to affordable cislunar transport, yet the basic engineering question (which oxygen-extraction route costs the least *energy* per kilogram of product) is hard to answer from the literature, because published figures are reported on incompatible bases: hydrogen reduction as electrical energy, carbothermal reduction as thermal energy, and molten regolith electrolysis (MRE) with no published figure at all. Prior work compares subsets (Taylor and Carrier (1993) reviewed ∼20 processes; Leger et al. (2025) modeled several on a common basis), but no published comparison places all five of these routes, including the thermally-reported carbothermal route and the two electrolysis routes that lack any published energy figure, on a single electrical-equivalent basis with paired uncertainty propagation. We present a small, open, uncertainty-quantified model that places five routes (hydrogen reduction, carbothermal reduction, MRE, molten-salt electrolysis, and polar water-ice mining) on one electrical-equivalent kWh/kg O_2 basis under a single explicit system boundary, with Monte-Carlo propagation of literature parameter ranges. We check the framework against the one route with a clean published electrical figure (hydrogen reduction, Leger et al. (2025), 24.3 ± 5.8 kWh/kg LOX): our nominal estimate (24.6) lands within ∼1% of Leger's central value, though our Monte-Carlo distribution is centered somewhat lower (median ∼21), so we report this as substantial interval overlap between two independent estimates rather than a point match. We add historical and terrestrial cross-checks for MRE (Carr 1963, as reported by Schreiner et al. 2016; terrestrial molten-oxide electrolysis), noting these span different system boundaries. Using a paired Monte Carlo that shares common parameters across routes, we report dominance probabilities rather than overlapping error bars. Three findings emerge: (1) carbothermal reduction is the most likely lowest-energy route (cheapest in 58% of paired trials; it beats hydrogen reduction ∼100% and MRE ∼90% of the time but only ∼75% versus molten-salt and the water route), visible only once routes share a common basis, though carbothermal is itself one of the routes without an independent energy anchor; (2) in 99% of trials the most energy-intensive route is either hydrogen reduction (51%) or MRE (48%), and MRE's reputation as low-energy does not survive a realistic full-cell voltage; (3) the polar water route, though mid-pack per kg O_2, is the only route that yields a complete LOX+LH_2 propellant. A sensitivity analysis identifies the single highest-value measurement for each route. We translate energy into required surface power and landed fission-plant mass, and outline two companion analyses on reusing co-located compute waste heat.

1 Introduction

The case for lunar propellant rests on energy: every process that turns regolith or polar ice into liquid oxygen (and ideally liquid hydrogen) is energy-intensive, and surface power is the binding constraint on any near-term architecture. Yet the comparison that should drive route selection has never been made on equal footing. Three incompatibilities recur in the literature:

1. **Thermal vs electrical energy are conflated.** Carbothermal reduction is reported as delivered thermal energy or "g O_2/kWh thermal"; hydrogen reduction is reported as electrical kWh/kg. On the Moon both ultimately draw on the same scarce electrical supply, but they are not interchangeable at face value.
2. **System boundaries differ.** Some figures include excavation, beneficiation, and liquefaction; others count only the reactor.
3. **Two routes have no published figure at all.** Molten regolith electrolysis and molten-salt electrolysis are described qualitatively; their energy cost is asserted, not quantified.

The consequence is that capital allocation, power-plant sizing, and architecture choice rest on a comparison that does not exist. This is a modeling gap, not a hardware gap, which is precisely why it can be closed now, cheaply, and reproducibly. We build that comparison and quantify its uncertainty.

2 Methods

2.1 Functional unit and system boundary

We compute the **electrical-equivalent energy in kWh per kg of O_2 delivered to cryogenic storage**, and additionally report kWh per kg of total propellant for the one route that co-produces hydrogen. The system boundary is a fixed sequence of stages; every route enables a subset, using the same stage sub-models, so all differences between routes arise from parameters rather than inconsistent accounting:

> excavation/acquisition → beneficiation → heating (sensible, plus fusion for melt routes) → reaction (reduction enthalpy or Faradaic electrolysis) → product cleanup → water electrolysis (where the route produces H_2O) → gas compression → liquefaction (LOX always; LH_2 where hydrogen is retained).

2.2 Thermal-to-electrical conversion

To compare a thermally-reported route against electrically-reported ones, all thermal demand is converted to electrical-equivalent through an explicit electric-to-thermal efficiency parameter (resistive heating, nominal 0.90). This single, visible knob is what makes the comparison honest; a solar-thermal heating pathway would be a separate sensitivity, not the baseline.

2.3 Uncertainty propagation

Every uncertain input is a triangular `(low, nominal, high)` distribution sourced from the literature (the full parameter table, with a citation on every value, is in `params.py`). A 20,000-trial Monte Carlo propagates these to a 90% interval per route. We propagate stated literature ranges, not assumed Gaussian measurement error. The electrolysis cell voltage and current efficiency of the electrochemical routes are sampled with a physical anti-correlation (a shared operating-severity latent: higher current density raises voltage and lowers efficiency together), avoiding unphysical parameter corners.

2.4 Ranking by paired Monte Carlo

Reading a ranking from five marginal error bars is a statistical error, because the routes share parameters (regolith specific heat, heat recuperation, electrolysis efficiency, liquefaction) that must take the same value in any given world. We therefore run a paired Monte Carlo: one shared parameter draw per trial across all routes, evaluated trial by trial, so we can report P(route A cheaper than route B) and P(route is cheapest / worst).

3 Validation

3.1 Hydrogen reduction against Leger 2025 (and a second anchor)

Hydrogen reduction is the only route with a clean published electrical figure: 24.3 ± 5.8 kWh/kg LOX, with the reduction step ∼55% and water electrolysis ∼38% of the total (Leger et al., 2025). We built our estimate bottom-up from first principles and independent parameters; the largest tunable term (water-electrolysis efficiency) is set from an independent SOEC source (Hauch et al., 2020; International Energy Agency, 2019), not from Leger's implied value.

- Nominal point estimate: **24.6 kWh/kg LOX**, within ∼1% of Leger's central value and inside his 1-sigma interval [18.5, 30.1].
- The Monte-Carlo distribution is centered lower (median ∼21, mean ∼22): Leger's value sits near our 85th percentile, not our center. We therefore report agreement as substantial **interval overlap** between two independent estimates, not as a point match; the nominal coincidence to ∼1% should not be over-read.
- Stage shares match: heating + reaction ∼65% (Leger ∼55%); water electrolysis ∼30% (Leger ∼38%).
- A second cross-check: Taylor and Carrier (1993) place the route at ∼26 kWh/kg LOX (cross-technology range 18–35).
- Because the model omits several one-signed terms (Section 9) that would raise totals, the nominal agreement with Leger and the lower distribution center are both consistent with the two estimates sharing a partial system boundary; we do not claim the absolute level is converged.

3.2 MRE against Carr 1963 and terrestrial molten-oxide electrolysis

MRE began as a pure first-principles estimate (nominal 18.5, median ∼21, 90% CI [12.9, 31.5]). Because it has no published electrical figure, we cross-check it against three sources that unavoidably use *different system boundaries*, so the agreement is weak and bounding rather than a clean anchor: Carr (1963), as reported secondarily by Schreiner et al. (2016), gives ∼26.4 kWh/kg O_2 for lunar MRE; terrestrial molten-oxide electrolysis of iron runs ∼3.7–4.0 MWh/t metal, i.e. ∼9 kWh/kg O_2, a well-insulated lower bound for a different chemistry (Allanore, 2015); and NASA reactor-sizing models place *whole-system* figures (including Joule heating and duty cycle, which our standalone-reactor boundary excludes) at ∼50–120 kWh/kg O_2 (Schreiner et al., 2016). Our estimate falls between these, which given their ∼9–120 spread is corroboration only in a loose sense. We did, however, correct the oxide-melt current efficiency to match *measured* regolith-surrogate data (70–90%, Ir anodes Allanore, 2015) rather than an optimistic assumption, a change that lowered MRE's estimate against the prior narrative.

These are the project's falsifiable tests (`tests/test_validation.py`). Only hydrogen reduction has a like-for-like published electrical figure; the others are first-principles estimates reported with wide intervals and as probabilities, not point claims.

4 Results

The single value is the deterministic nominal (all parameters at their cited value). For the right-skewed electrochemical routes the Monte-Carlo *median* differs materially: notably MRE and hydrogen reduction share a median of ∼21 kWh/kg O_2, so although their nominals differ (18.5 vs 24.6) they are statistically tied as the two most intensive routes (Table 2). For plant sizing, prefer the median and the interval over the nominal.

Table 1: Electrical-equivalent energy per route (20,000 trials).

Route	Yields	kWh/kg O_2 (nominal)	90% CI	kWh/kg propellant
Carbothermal (CH_4)	LOX	13.4	12.3–15.5	13.4
PSR water mining	LOX+LH_2	14.4	12.8–17.3	**12.8**
Molten-salt (FFC)	LOX	15.3	12.2–21.6	15.3
Molten regolith electrolysis	LOX	18.5 (median 21)	12.9–31.5	18.5
H_2 reduction (ilmenite)	LOX	24.6 (median 21)	16.9–27.6	24.6

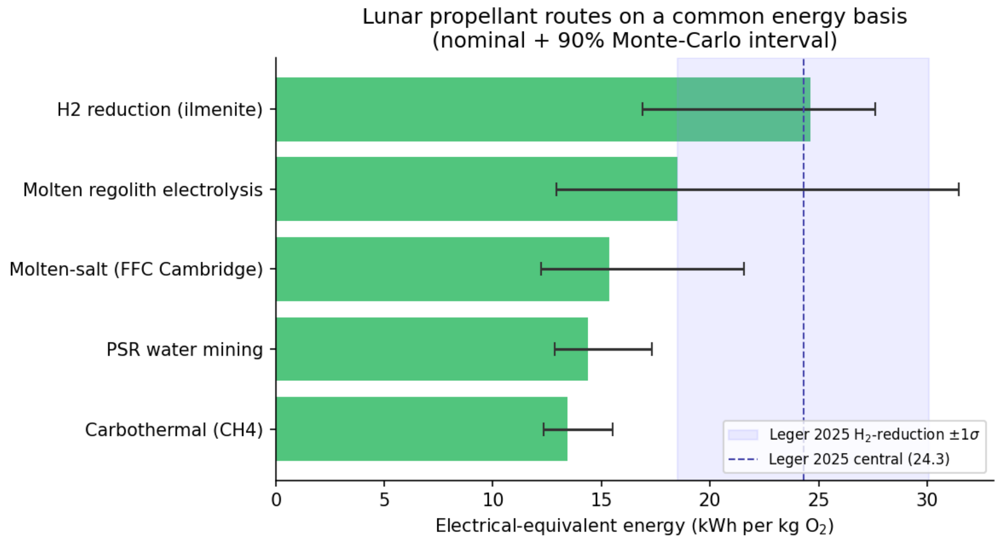

Figure 1: The five routes on a common electrical-energy basis (nominal + 90% Monte-Carlo interval); the shaded band is Leger 2025's 1-sigma for hydrogen reduction.

Carbothermal beats hydrogen reduction in 100% of paired trials and MRE in ∼90%. Molten-salt and the water route trade second place and are not separable.

5 Sensitivity analysis

A one-at-a-time tornado analysis (each parameter swept low-to-high, all others nominal; `python -m lpem -sensitivity <route>`) identifies the dominant uncertainty for each route and, with it, the single highest-value measurement:

Two practical conclusions: MRE's entire energy verdict hinges on one unmeasured number, its full-cell decomposition voltage; pinning it with reactor data is the highest-value measurement this model identifies. The water route's verdict hinges on small-scale LH_2 liquefaction performance. The route ranking itself is robust: carbothermal's lead and the hydrogen-reduction/MRE disadvantage survive across the parameter ranges (Section 6).

6 Findings

Finding 1: Carbothermal is the most likely lowest-energy route, visible only on a common basis. Cheapest in 58% of paired trials; it beats hydrogen reduction in ∼100% of trials and MRE in ∼90%, but only ∼75% versus molten-salt and the water route, so it is the most likely winner, not a certain one. Its high O_2 yield (>20 wt%) means it heats only ∼5 kg of regolith per kg O_2, versus ∼50 kg for hydrogen reduction. Recast from its usual thermal basis

Table 2: Paired-Monte-Carlo dominance.

Route	P(cheapest)	P(worst)
Carbothermal (CH_4)	0.58	0.00
Molten-salt (FFC)	0.22	0.00
PSR water mining	0.19	0.00
Molten regolith electrolysis	0.02	0.48
H_2 reduction (ilmenite)	0.00	0.51

Table 3: Dominant sensitivity driver per route.

Route	Dominant driver (swing, kWh/kg O_2)	Next
H_2 reduction	O_2 yield (13.3)	heat recuperation (9.1), cp (5.3)
Carbothermal	reaction enthalpy (3.5)	yield (1.8), electrolysis eff (1.7); flat tornado
MRE	**cell voltage (15.9)**	yield (3.8), current efficiency (3.7)
Molten-salt	current efficiency (5.6)	cell voltage (4.5)
PSR water	**LH_2 liquefaction (6.0)**	electrolysis eff (1.7), ice grade (1.0)

onto the common electrical basis, it comes out ahead, a comparison not previously made. One caveat we state plainly: unlike hydrogen reduction and MRE, carbothermal has no independent published energy figure, and its lead rests on an assumed high yield and a first-principles reaction enthalpy that includes a Sabatier exotherm credit. Its lead is robust to its *own* parameters (flat tornado, Section 5) but is model-internal pending a measured electrical figure.

Finding 2: Hydrogen reduction and MRE are the two most energy-intensive, and MRE's "low energy" reputation does not survive a realistic cell voltage. On the Monte-Carlo median they are tied (~21 kWh/kg O_2 each) and one of the two is the worst route in 99% of trials (hydrogen reduction 51%, MRE 48%). Hydrogen reduction pays to heat ~50 kg regolith per kg O_2 at low ilmenite yield plus a separate electrolysis step; it is reliably poor. MRE is the high-variance one (median ~21, CI [12.9, 31.5]): its Faradaic term depends on the full decomposition voltage (~3.5 V including overpotential and ohmic drop), not the ~1.7 V thermodynamic floor, and not the 16–34 V Joule-*heating* voltages sometimes quoted from concept papers. With measured current efficiency, MRE is the single worst route whenever its cell voltage runs high.

Finding 3: Per-kg-O_2 scoring understates the only full-propellant route. Polar water mining is the only route that co-produces usable hydrogen, so it alone yields a complete LOX+LH_2 propellant rather than LOX with fuel still shipped from Earth. It is mid-pack per kg O_2 (14.4) but its value is closing the hydrogen loop, not lowest energy; the apparent per-propellant edge (12.8) is within uncertainty and excludes small-scale LH_2 liquefaction and zero-boil-off storage.

7 From energy to power plant and landed mass

Energy matters through what it costs to supply. Converting kWh/kg into continuous surface power and the landed mass of the fission-surface-power (FSP) system it implies (`lpem.arch`), even a modest 50 t O_2/yr plant requires ~85–156 kWe, comparable to NASA's 40-kWe FSP concept (Oleson et al., 2022) scaled up (and roughly one to ~1.6 units of the later 100-kWe FY2030 target). Route choice swings landed power-system mass by ~16 t (carbothermal ~19 t vs hydrogen reduction ~35 t) for the same oxygen output. The direction is robust; the magnitude scales with the FSP specific mass and assumes an all-fission architecture (a solar-plus-storage plant would be dominated by night-survival storage mass, not modeled here).

8 Companion analyses (compute waste-heat integration)

Two companion analyses, kept modular, examine reusing co-located compute waste heat:

- **Low-grade thermal offset** (`WASTE-HEAT-OFFSET.md`). By the Second Law, compute waste heat (∼330 K) can supply only low-grade ISRU demand. At the nominal 350 K reject temperature, comfortably above the ∼273 K sublimation target, it can in principle cover the water route's full thermal-mining term (∼2.1 kWh/kg O_2, ∼14% of the route; Figure 2); this offset falls off sharply and excludes the sublimation enthalpy if compute rejects below ∼273 K. It cannot supply high-grade reduction/electrolysis heat at all.
- **PSR co-location** (`PSR-COLOCATION.md`). A permanently shadowed region is both a cryogenic radiative sink for compute and the location of the water resource. Under an explicit radiator energy balance, PSR siting saves a median ∼46 t of radiator mass per MW of compute (nominal 51; 90% CI ∼11–290, the wide upper tail reflecting and heuristically capping sunlit cases that cannot reject at all; Figure 3), and in ∼18% of sampled conditions a sunlit 330 K radiator cannot reject at 330 K without active cooling. The heat cascade itself is a modest bonus (saves ∼2.7 t reactor for the reference plant, break-even enabling probability ∼38%); co-location is justified by the compute siting economics, not the cascade.

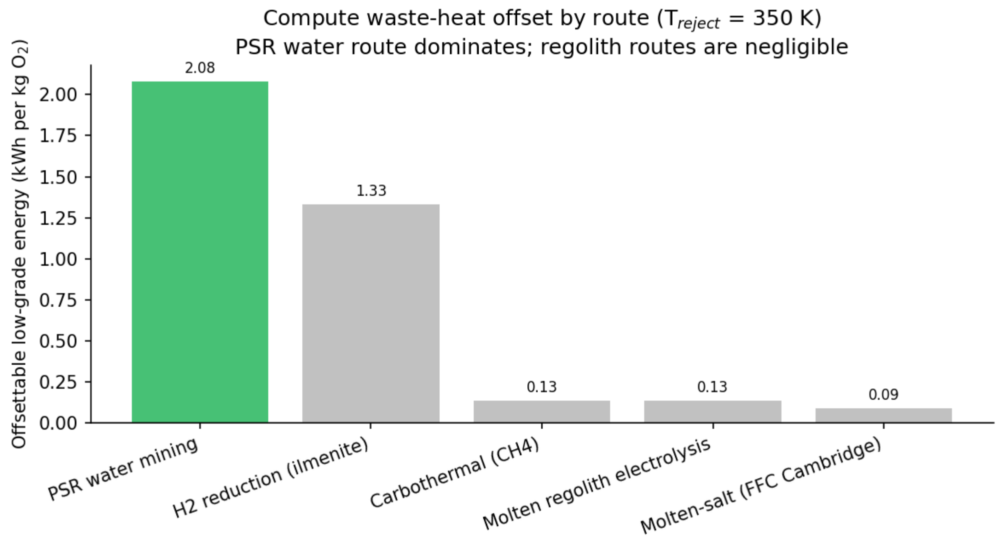

Figure 2: Low-grade, waste-heat-offsettable energy per route ($T_{\text{reject}} = 350$ K).

9 Limitations

- Steady-state production energy, power, and FSP landed mass only; capital cost, mobility, comms, site logistics, and demand are out of scope.
- Omitted energy terms are one-signed (they raise totals): standing radiative loss from continuously hot reactors (worst for the hottest route, MRE), comminution/beneficiation, electrolyzer heat-rejection parasitics, and, for the water route, months-long LH_2 zero-boil-off and power delivery into permanent shadow. Absolute kWh/kg figures are best read as lower bounds; the ranking is more robust than the levels.

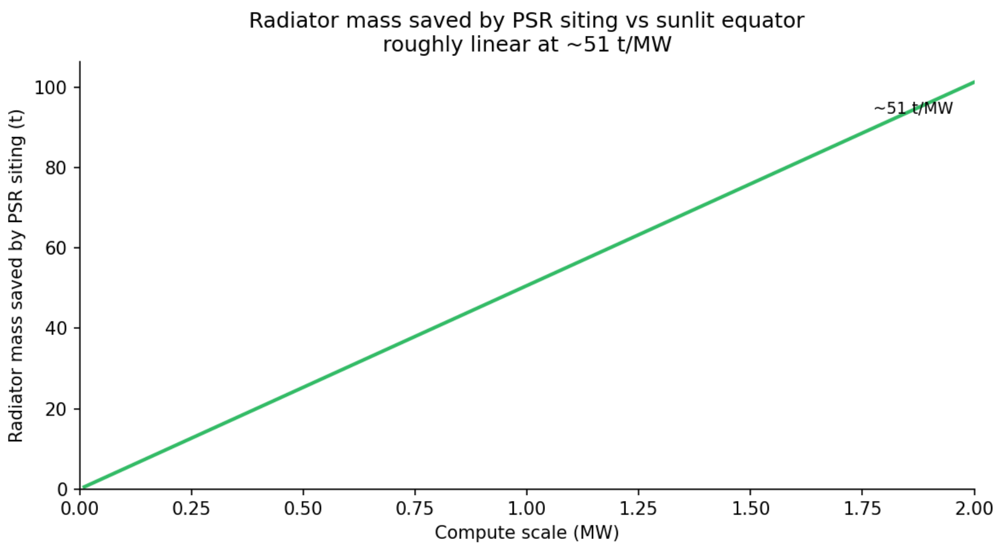

Figure 3: Radiator mass saved by PSR siting vs a sunlit site (nominal ∼51 t/MW).

- Three of five routes lack a direct independent electrical anchor.
- O_2 yield and reaction temperature are sampled independently (a residual idealization).
- Site geography and solar-thermal process heat are not modeled.

10 Reproducibility

```
pip install -e .
python -m lpem                    # Table 1
python -m lpem --dominance        # Table 2
python -m lpem --sensitivity mre    # Section 5 tornado
python -m lpem --plant-tonnes 50    # Section 7
python -m lpem --waste-heat --benefit    # Section 8
python scripts/make_figures.py      # Figures 1-3
pytest                            # 48 tests, incl. the validation anchors
```

References

The full bibliography (with DOIs/URLs) is in `paper/REFERENCES.md` and `paper/references.bib`. Key sources: Leger et al. (2025) (PNAS); Taylor and Carrier (1993); Carr (1963); Allanore (2015) (J. Electrochem. Soc.); Schreiner et al. (2016) (Adv. Space Res.); Kornuta et al. (2019) (REACH); Hauch et al. (2020) (Science); Colaprete et al. (2010) (Science).

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